Chapter 12
Trading Strategies Involving Options

Options can be combined into an unlimited number of payoff shapes, and textbooks that catalog them tend to leave readers able to draw a great many diagrams and unable to say why anyone would want one. This chapter is organized around the opposite question. Every position is a bet on something, and a well-chosen strategy is one whose bet matches the view its holder actually has.

There are only four things to bet on: **direction**, where the underlying goes; **magnitude**, how far it moves, regardless of which way; **time**, whether the move happens before the option expires; and **skew**, the relative price of downside and upside protection. Most strategies are a bet on two or three of these simultaneously, which is how a trader can lose money on a position whose directional view turns out to be right.

Every one of these bets is levered. A spread that costs $2.04 on a $48.30 stock earns up to 1.45 times its cost if the stock rises 8.7%, and loses all of it if the stock falls 1.7%. Leverage is not a fifth thing to bet on; it is the multiplier on the other four, the same multiplier Example 1.5 measured for futures, and it is why a position whose diagram looks tame can be anything but tame in a portfolio. The examples below therefore report what each position costs alongside what it can pay, because the ratio of the two is where the leverage sits.

Chapter 11 has already supplied the tools. Put–call parity means many strategies that appear distinct are the same position in a different form, and Section 12.1 begins there because it removes much apparent complexity at the outset. Sections 12.2 to 12.6 take the strategies by the view they express, from spreads for direction to risk reversals for skew; the rest of the chapter prices one view four ways, builds arbitrary payoffs, turns to the institutional uses (the collar and the portfolio hedge), sums up what each strategy bets on, and closes with structured products, position management and costs.

In the worked examples, the numbers come from the option chain of Table 10.2 (a stock at $48.30, 45 days to expiration), and they are computed at the *bid and ask*, not the mid. Section 12.14 shows why that decision changes the conclusions.

12.1 STRATEGIES THAT ARE EQUIVALENT

Two of the most widely used retail strategies are not new positions at all. A **covered call** is long stock plus a written call. Parity (11.4), which holds exactly for European options, rearranges to

$$S_0 - c = Ke^{-rT} - p$$

so the covered call has the payoff of a *written put* plus cash. A **protective put** is long stock plus a purchased put, and the same rearrangement gives

$$S_0 + p = c + Ke^{-rT}$$

which is a *purchased call* plus cash. Figure 12.1 shows both.

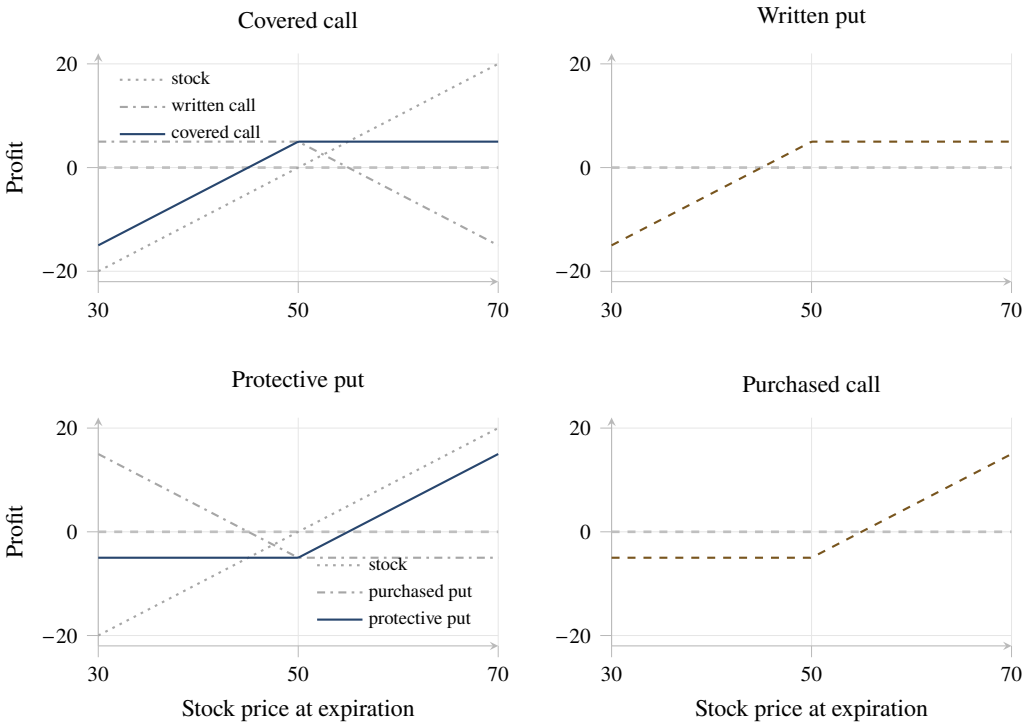


Figure 12.1: Profit at expiration of each strategy, with its components in gray, and of its parity equivalent. $K = 50$.

The panels in each row are identical, and this is not a curiosity: it has three practical consequences that recur through the chapter.

It disciplines the marketing. A covered call is often sold as a conservative income strategy and a written put as a speculative one. Held to expiration, the covered call is a written put plus cash, a **cash-secured put**, and brokers, who set options-approval levels, commonly approve the two at the same lowest level; the higher level is for the *naked* put, the same exposure without the cash. The *market risk* is identical.

It tells you which to trade. If two routes to the same exposure exist, take the cheaper. Writing the put is one transaction rather than two, so it pays the bid–ask spread once instead of twice.

It identifies the real bet. A covered call writer is short a put: they are selling insurance, and they profit when the market is calm and lose when it falls. Describing this as "generating income from a stock you already own" is accurate about the cash flow and silent about the risk.

12.2 SPREADS: A DIRECTIONAL VIEW WITH BOTH ENDS CAPPED

A **spread** combines two or more options of the same type, each of them a **leg** of the spread. The simplest is a directional bet financed by selling away the part of the move you do not expect.

A **bull call spread** buys a call and sells a higher-strike call. It costs less than the call alone, because the sale pays for part of it, and in exchange the profit stops at the higher strike. A **bear put spread** is the mirror image (Example 12.2). Each can also be built from the other type of option: a **bull put spread** sells a put and buys a lower-strike put, and a **bear call spread** sells a call and buys a higher-strike call. These bring in a net premium, so for each view one

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version is a **debit** spread, which costs a net premium to establish, and the other a **credit** spread. By put–call parity the two versions of the same view have the same profit diagram apart from interest on the strike difference, so the choice between them is about cash up front and margin, not about the view; put credit spreads are a staple of investors who sell options for income.

Example 12.1 A bull call spread

Using Table 10.2, buy the 47.50 call at its \$2.72 ask and sell the 52.50 call at its \$0.68 bid:

$$\text{net debit} = 2.72 - 0.68 = \$2.04 \text{ per share, or } \$204 \text{ per spread}$$

The outcomes are:

Stock at expiration	Payoff	Profit
≤ 47.50	0.00	−2.04
49.54	2.04	0.00
≥ 52.50	5.00	+2.96

The maximum profit is \$2.96, the maximum loss \$2.04, and the breakeven \$49.54. The stock must rise 2.6% in 45 days merely to break even, and 8.7% to earn the maximum. The **reward-to-risk ratio** (maximum profit divided by maximum loss, a trader's shorthand that says nothing about how likely either is) is 1.45 to 1.

Compare with buying the 47.50 call outright at \$2.72. That costs 33% more and has unlimited upside. The spread does better at every price below \$53.18, rise or fall, and the outright call above it. The spread is not a cheaper way to be bullish; it is a bet on a specific range.

Example 12.2 A bear put spread

This is the mirror image, for a trader who expects a fall. Using Table 10.2, buy the 50 put at its \$3.95 ask and sell the 45 put at its \$1.81 bid:

$$\text{net debit} = 3.95 - 1.81 = \$2.14 \text{ per share, or } \$214 \text{ per spread}$$

The outcomes are:

Stock at expiration	Payoff	Profit
≥ 50	0.00	−2.14
47.86	2.14	0.00
≤ 45	5.00	+2.86

The maximum profit is \$2.86, the maximum loss \$2.14, and the breakeven \$47.86. The stock must fall 0.9% in 45 days to break even and 6.8% to earn the maximum, and the reward-to-risk ratio is 1.34 to 1, against 1.45 to 1 for the bull spread of Example 12.1. Part of the gap is moneyness (the bull spread buys a strike \$0.80 below spot, the bear spread one \$1.70 above) and part is the chain's dividend; Example 12.5 separates the forward from the skew.

12.3 BUTTERFLIES AND CONDORS: A VIEW THAT THE PRICE STAYS PUT

Section 11.3 introduced the butterfly as the position whose value convexity keeps nonnegative. As a trade, it profits if the underlying stays near the middle strike until expiration.

Buy one call at K_1, sell two at K_2, buy one at K_3, with equal spacing. The payoff is zero outside $[K_1, K_3]$ and peaks at $K_2 - K_1$ in the middle. A **condor** spreads the two middle strikes apart, producing a plateau instead of a peak: a wider target for a lower maximum. Most condors are traded as **iron condors**: an out-of-the-money put spread and an out-of-the-money call spread, both sold, which brings in a credit and, by parity, gives the call condor's profit profile. This credit form is what premium-selling funds run, and Chapter 23 finds such premium selling

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(put spreads, call spreads and iron condors) a large and growing share of same-day index options volume. Figure 12.2 collects the four profit profiles, which together show what every spread buys and sells: the strike that was sold caps the profit, and the premium it brings in is what makes the structure cheaper than the option bought alone.

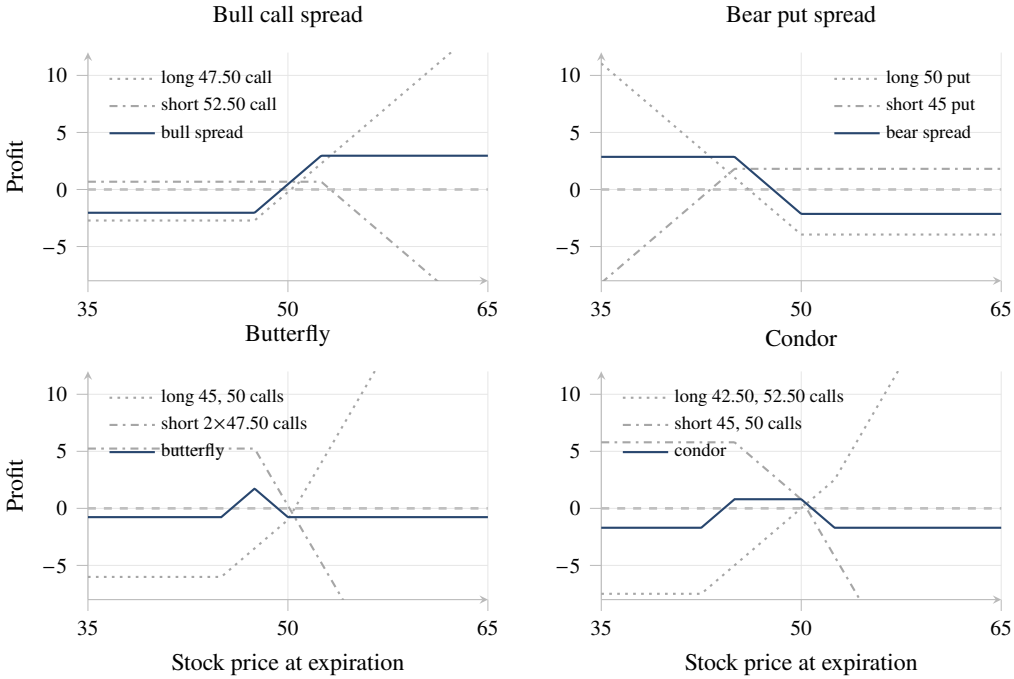


Figure 12.2: Profit at expiration for four spread structures at the quotes of Table 10.2, with the legs in gray.

Example 12.3 A butterfly, priced at the quotes

From Table 10.2, buy the 45 call at \$4.50, sell two 47.50 calls at \$2.62, buy the 50 call at \$1.51:

$$\text{cost} = 4.50 - 2 \times 2.62 + 1.51 = 6.01 - 5.24 = \$0.77$$

The maximum payoff is \$2.50, achieved if the stock finishes at exactly \$47.50, so the maximum profit is \$1.73 against a maximum loss of \$0.77, a reward-to-risk ratio of 2.25 to 1. The breakevens are \$45.77 and \$49.23.

That ratio looks attractive until the window is examined. The stock is at \$48.30, so it must finish between 5.2% below and 1.9% above its current level, in 45 days, for the position to make anything at all. Butterflies pay well precisely because that is unlikely.

Now comes the point Section 11.3 set up. At mid prices Example 11.2 found the same butterfly worth \$0.56, so the trader paid \$0.77 for something worth \$0.56: execution consumed \$0.21 (27% of the price paid, and 38% of the position's theoretical value) before the market moved at all. Section 12.14 shows that this cost recurs in every multi-leg strategy.

12.4 STRADDLES AND STRANGLES: A VIEW ON THE SIZE OF THE MOVE

To bet that something will happen without a view on which way, buy both a call and a put. A **straddle** uses the same strike; a **strangle** uses a lower put strike and a higher call strike, which is cheaper and needs a move at least as large to pay.

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Example 12.4 How far the stock must move

Buy the 47.50 straddle at the asks: call \$2.72 plus put \$2.63 equals \$5.35.

The breakevens are 47.50 ± 5.35, that is \$42.15 and \$52.85. From the current \$48.30 the stock must fall 12.7% or rise 9.4% within 45 days for the position to make a cent.

A strangle using the 45 put (\$1.89 ask) and the 50 call (\$1.51 ask) costs \$3.40, 36% less, but its breakevens are \$41.60 and \$53.40, outside the straddle's on both sides. The strangle costs less and risks less, but it needs a larger move, because the market has priced both structures off the same volatility surface.

This is the general result. An option position's cost and the move it requires are two views of one number, the implied volatility. Buying a straddle is not a bet that the stock will move; the market already expects it to. It is a bet that the stock will move *more than the implied volatility says*—and Chapter 19 documents that implied volatility exceeds subsequent realized volatility on average. The straddle buyer is systematically paying a premium for that bet.

Figure 12.3 draws the two positions of Example 12.4 side by side.

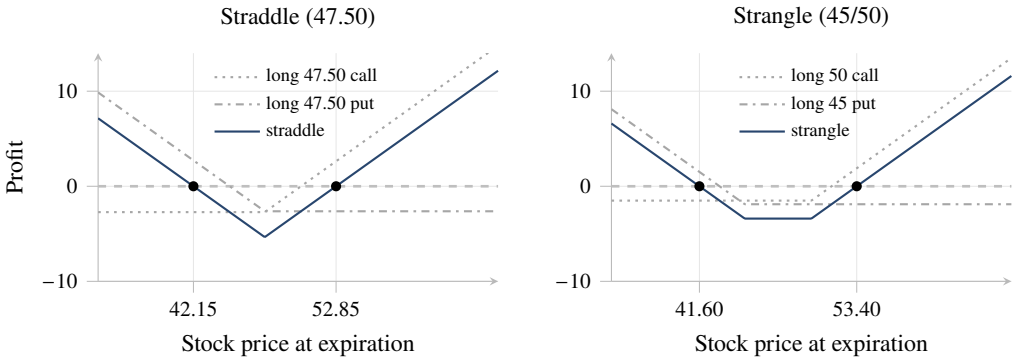


Figure 12.3: Profit at expiration for the straddle and strangle of Example 12.4, at the quotes of Table 10.2, with the legs in gray.

The raw straddle is not a pure volatility trade. An at-the-money straddle has almost no exposure to the direction of a small move, because the call gains about as much as the put loses. That balance holds only while the stock stays near the strike. If the stock rises, the call moves into the money and comes to behave more like the stock, while the put moves out of the money and matters less, so the position as a whole now gains when the stock rises and loses when it falls: it has become net long. Chapter 18 measures that sensitivity as the option's delta and shows why it changes with the stock price. The consequence needs no measurement to see: a straddle held through a large move ends up as a directional position in whichever direction the stock happened to go.

A trader who genuinely wants only volatility exposure must **delta-hedge**: trade the underlying continuously to keep the net directional exposure at zero, so that what remains is a bet on realized against implied volatility alone. Chapters 18 and 22 develop the trade: Chapter 18 measures the gamma (how fast the delta changes as the stock moves) and the theta (the value lost each day to time decay) that the hedge trades against each other, and Section 22.1 shows that the hedging profit accumulated along the way is the difference between realized and implied variance, weighted by gamma. The unhedged straddle of Example 12.4 is a coarser instrument: it contains the volatility bet, plus a directional bet the trader did not intend and is not being paid for.

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12.5 CALENDAR SPREADS AND THE TERM STRUCTURE

A **calendar spread** sells a near-dated option and buys a longer-dated one at the same strike. It is not a directional trade and not straightforwardly a volatility trade; it is a bet on two things at once.

The first is **time decay**. A near-dated option loses time value faster than a long-dated one: Section 11.1 noted that value grows roughly with $\sqrt{T}$, so the decay per day is steepest at the end. Selling the fast-decaying option against the slow one collects that difference, provided the stock stays near the strike. Figure 12.4 shows both halves of the argument: the left panel is the spread's profit at the near expiration, a hump that peaks at the strike and falls away on either side, and the right panel is time value against remaining life, with the near-dated option sitting on the steep part of the curve.

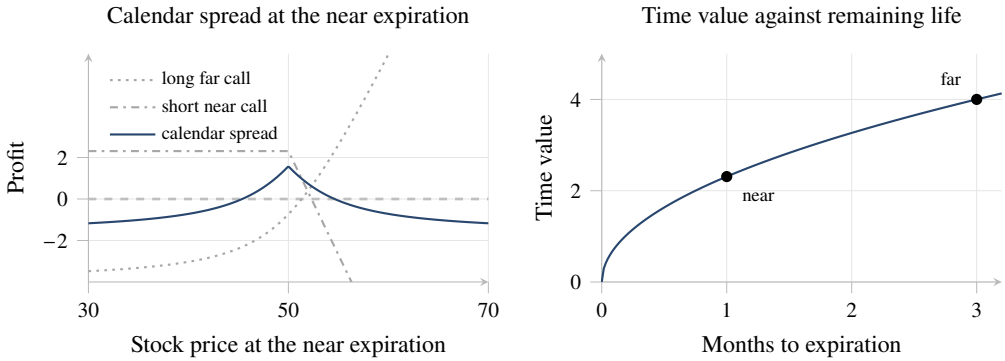


Figure 12.4: Left: profit on a calendar spread at the near option's expiration, $K = 50$, schematic. Right: time value against remaining life, proportional to $\sqrt{T}$.

The second is the **term structure of implied volatility**. The two legs are usually priced at different implied volatilities, and the calendar spread gains if the far option's implied volatility rises and loses if the near one's does. When near-dated implied volatility is elevated (before an earnings announcement, say), selling the near option at that level against a far-dated option priced at a calmer one is the trade the structure is designed for. Chapter 19 treats that term structure properly.

A **diagonal spread** changes the strike as well as the maturity, combining the calendar bet with a directional one. It is popular and frequently misunderstood, because its holder often has a clear view on only one of the two bets they have taken.

12.6 RISK REVERSALS AND RATIO SPREADS: A VIEW ON SKEW

A **risk reversal** buys an out-of-the-money call and sells an out-of-the-money put. Section 11.5 showed that at the same strike this is a synthetic long; at different strikes it is a long position with a gap in the middle, and it can often be established for zero cost.

What makes it interesting is what determines that cost. Buying the call and selling the put trades one implied volatility against another, so the price of a risk reversal *is* the skew: the amount by which downside protection is more expensive than upside participation. In equity markets the lower-strike put almost always carries the higher implied volatility, because puts are persistently in demand, so the trader who buys the call and sells the put is *paid* the skew in volatility terms: the option sold is the dear one and the option bought the cheap one. In dollars the risk reversal of Section 12.7 is entered for a small credit, and since its two strikes sit about equally far from the forward, that credit is mostly the skew. Chapter 19 explains why it is there.

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Risk reversals are therefore the standard way to trade a view on skew rather than on the level of volatility. They are also the standard corporate hedge: a firm wanting downside protection at zero premium sells away the upside to pay for it, which is a collar (a put bought and a call sold against a holding; Section 12.9) and is the same structure viewed from the other side. That side pays the skew; Examples 12.5 and 12.6 measure what it costs.

A **ratio spread** buys one option and sells more than one further out. It is cheap to establish, sometimes for a credit, and profits over a range, but the extra short options leave the position exposed beyond them. The shape looks safe on a profit diagram and is not, because the loss is unbounded (for calls; very large for puts) in exactly the tail the trader expects not to occur. Figure 12.5 shows both: the risk reversal is a long position with a flat stretch between the strikes, and the ratio spread's profit turns down without limit beyond the strike it sold twice.

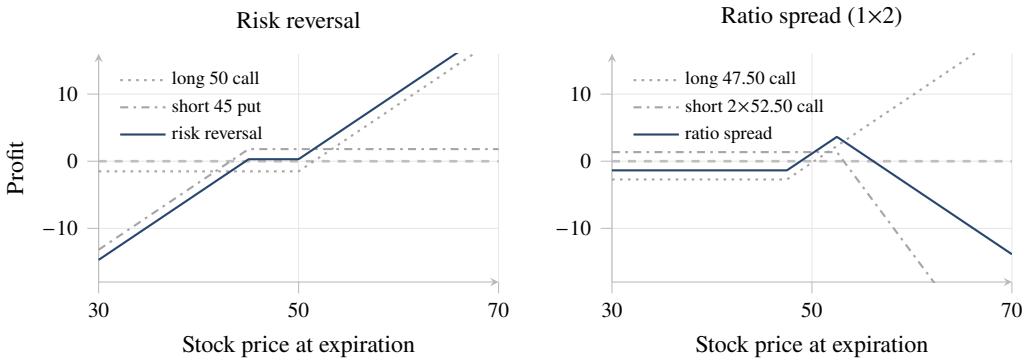


Figure 12.5: Profit at expiration for a risk reversal and a 1×2 ratio spread at the quotes of Table 10.2, with the legs in gray.

12.7 EXPRESSING ONE VIEW FOUR WAYS

The abstract discussion becomes concrete when a single view is priced through several instruments. Suppose a trader believes the stock, at \$48.30, will rise to around \$52 within 45 days. Table 12.1 shows what happens under four ways of acting on that: buying the stock, buying the 47.50 call outright, the 47.50/52.50 bull spread of Section 12.2, and a risk reversal. The outright call is the spread's own long leg, so the call and spread columns differ only by the call the spread sold.

The risk reversal here buys the 50 call at \$1.51 and sells the 45 put at \$1.81, for a net credit of \$0.30. Its capital line is not that credit. The written put must be margined, and formula (a) of Section 10.8 applied to it (20% of \$48.30, less the \$3.30 by which the put is out of the money, plus the \$1.81 premium) requires \$8.17 a share, so the route commits $817 - 30 = \$787$. Four things in the table are worth extracting.

The structure tailored to the view wins when the view is exactly right. At \$52, the trader's actual target, the bull spread earns \$246, more than either other option route, and returns 121% on its capital, more than any route; only the stock earns more in dollars, \$370, with twenty-four times the capital. This is what it means for a strategy to match a view.

It loses when the view is right about direction and wrong about size. At \$56 the spread earns \$296 while the outright call earns \$578. The spread sold that outcome away, which is precisely how it was made cheap.

Zero cost is not zero risk. The risk reversal was entered for a \$30 credit and still loses \$270 at \$42, almost as much as the outright call, which cost \$272. Its cheapness comes from the written put, and the written put is the whole downside. A structure that costs nothing has not

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Table 12.1: Profit per 100 shares of exposure, at the quotes of Table 10.2, and the return on the capital each route commits.

Stock at expiration	Buy stock	47.50 call	47.50/52.50 spread	Risk reversal
Profit, $				
$42.00	−630	−272	−204	−270
$45.00	−330	−272	−204	+30
$48.30	0	−192	−124	+30
$52.00	+370	+178	+246	+230
$56.00	+770	+578	+296	+630
Capital committed	$4,830	$272	$204	$787
Return on capital committed				
$42.00	−13.0%	−100%	−100%	−34.3%
$45.00	−6.8%	−100%	−100%	+3.8%
$48.30	0	−70.6%	−60.8%	+3.8%
$52.00	+7.7%	+65.4%	+120.6%	+29.2%
$56.00	+15.9%	+212.5%	+145.1%	+80.1%

eliminated risk; it has financed one exposure by selling another.

Capital efficiency and risk run in opposite directions. Buying the stock ties up $4,830 and loses $630 in the bad case; the spread ties up $204 and loses $204. The return block makes the leverage explicit: at the target the spread returns 121% on its capital and the stock 7.7%, and at $42 the same spread returns −100% while the stock loses 13%. A return on capital is meaningless unless the loss column is read at the same time, which is the point Example 1.5 made about leverage, appearing here as a choice among strategies.

12.8 BUILDING AN ARBITRARY PAYOFF

The structures above are conventional, but nothing restricts a trader to them. A general result underlies the whole subject: it explains where exotic payoffs come from and is the bridge to Chapters 19, 22 and 34.

Any piecewise-linear payoff can be replicated exactly by a bond, a forward, and a portfolio of European calls. The recipe is mechanical: the bond and the forward match the payoff's level and slope at the lowest prices, and a call at each kink, in a quantity equal to the *change in slope* there, does the rest.

Take the bull call spread. Its payoff is flat at zero below $47.50, rises with slope 1 to $52.50, and is flat above. The slope changes by +1 at 47.50 and by −1 at 52.50, so the replicating portfolio is one long call at 47.50 and one short call at 52.50—which is what a bull spread is. The butterfly's slopes are $0, +1, -1, 0$, giving changes of $+1, -2, +1$: long one, short two, long one.

The same recipe builds anything. A note paying 70% of an index's rise, capped once the index has risen 30% (a maximum return of 21%), is a payoff with slope changes of +0.7 at the initial level and −0.7 at 130% of it: buy 0.7 calls at the first strike, sell 0.7 at the second. Figure 12.6 shows the construction.

Digitals as the limit of a spread. A **digital** (or binary) option pays a fixed amount if the underlying finishes above a strike and nothing otherwise. Its payoff has a jump rather than a kink, so no finite portfolio of calls reproduces it, but a tight call spread approaches it. Buying a call at $K - \varepsilon$ and selling one at $K + \varepsilon$, scaled by $1/(2\varepsilon)$, gives a payoff that rises from 0 to 1

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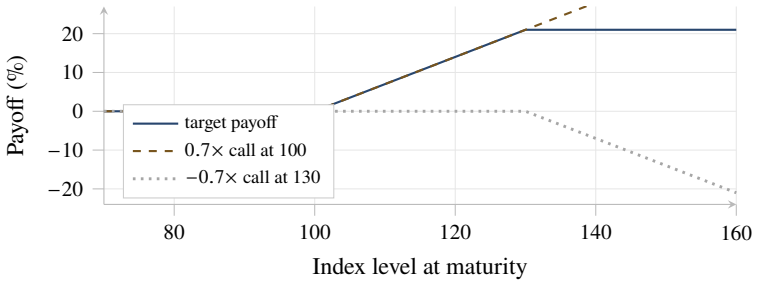


Figure 12.6: A capped participation payoff as two call positions.

across a narrow band. As $\varepsilon \to 0$ the position becomes a digital, and its price becomes

$$\text{digital}(K) = -\frac{\partial c}{\partial K}$$

This is not merely a pricing device; it is how digitals are hedged. A dealer short a digital holds a tight call spread, and the tighter the spread the larger the position required, so digitals are risky to hedge near the strike at expiration: the same pin risk of Example 10.6 in a more acute form. The autocallable coupons of Section 12.12 are digitals, so those products are difficult to hedge precisely when the index sits near the autocall trigger. Chapter 19 pushes the same construction one derivative further, to a butterfly of vanishing width, and reads the market's probability distribution for the underlying off the option chain.

12.9 COLLARS AND THE CONCENTRATED SHAREHOLDER

The most common institutional use of an option combination is not speculative. A founder, executive, or family office holding a large position in a single stock wants to reduce the risk without selling, because selling would trigger tax, breach a lockup (an agreement not to sell for a set period, typical after a public offering), or signal something unwanted.

The **collar** buys a put and sells a call to pay for it. Section 12.6 noted it is a risk reversal seen from the other side; here it is a hedge rather than a view.

Example 12.5 A collar, and why its strikes are not symmetric

A shareholder holds 200,000 shares at \$48.30, worth \$9.66 million. Using Table 10.2, they buy the 45 put at its \$1.89 ask and sell the 50 call at its \$1.44 bid:

$$\text{net} = 1.44 - 1.89 = -\$0.45 \text{ per share, a debit of } \$90{,}000$$

The call pays for three-quarters of the put. Leaving that \$90,000 aside, the position is floored at \$45 and capped at \$50:

$$\text{worst case} = 200{,}000 \times 45 = \$9.0\text{m}, \qquad \text{best case} = 200{,}000 \times 50 = \$10.0\text{m}$$

The put is 6.8% below spot; the call is only 3.5% above it. The holder gave up 3.5% of upside to protect against everything below 6.8% down, and still paid.

Two things impose that asymmetry. The first is the forward: the dividend Example 10.3 found in the chain puts the 45-day forward near \$47.60, from which the strikes are nearly symmetric, 5.5% below and 5.1% above, so on this chain the forward sets most of the gap in the strikes. The second is skew: measured from the forward the two options are nearly equidistant, yet the put costs \$0.38 more at mid (\$0.45 at the quotes), because the chain prices it at a 46% implied volatility against the call's 37% (Section 15.11). A zero-cost collar must therefore set its put further from the forward than its call, as Example 12.6 shows on index quotes. Chapter 19 explains why, and the reason connects directly to who is on each side: shareholders like this one persistently buy downside protection, and someone must be paid to supply it.

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Because skew is part of what sets the terms, a collar is cheapest to establish when skew is flat.

12.10 HEDGING A PORTFOLIO WITH INDEX OPTIONS

Section 3.6 hedged an equity portfolio with index futures. Options offer an alternative with a different shape, and the choice between them is the clearest institutional application of everything in this chapter.

The contract count is the same beta-adjusted calculation as for futures:

$$N = \beta \frac{V_P}{S \times \text{multiplier}} \tag{12.1}$$

where V_P is the value of the portfolio, β its beta against the index, S the index level, and $S \times$ multiplier the value of the index underlying one option contract (the role V_F played for a futures contract in Section 3.6). What differs is that a put costs money but can lose no more than that, while a futures hedge is free to establish and removes the upside along with the downside.

Example 12.6 What protecting a portfolio actually costs

A fund holds \$50 million of equities with a beta of 1.1 against an index at 5,250. The index option multiplier is \$100. From (12.1),

$$N = 1.1 \times \frac{50{,}000{,}000}{5{,}250 \times 100} = 1.1 \times 95.24 = 104.8 \rightarrow 105 \text{ contracts}$$

The fund buys three-month puts 5% out of the money, struck at 4,987.5 and quoted at 60.00 points:

$$\text{cost} = 60 \times 100 \times 105 = \$630{,}000 = 1.26\% \text{ of the portfolio, per quarter}$$

The floor is set on the index, not on the portfolio. With a beta of 1.1 a 5% fall in the index is roughly a 5.5% fall in the portfolio, and the beta-adjusted contract count is what makes the puts pay enough to match it. The beta scaling is an approximation that ignores interest and dividends; Section 16.2 derives the strike that corrects for them. If the index falls 10%, to 4,725, the portfolio loses about \$5.5 million while the 105 puts pay $262.5 \times 100 \times 105 = \$2.76$ million, so the loss net of the put payoff is \$2.74 million, or 5.5% of the portfolio, 6.7% once the \$630,000 premium is counted. The floor stays near 5.5% however far the index falls beyond the strike, drifting only with the rounding of 104.8 contracts to 105.

Renewed every quarter, the premium comes to roughly 5% a year. Set that against a long-run expected equity return in the high single digits and the conclusion is uncomfortable but unavoidable: continuous protection at a strike this close costs most of the expected return. This single number explains why most institutional hedging structures look the way they do.

The put-spread collar is the usual response. Buy the 4,987.5 put at 60, sell a 4,462.5 put (15% down) at 22, and sell a 5,512.5 call (5% up) at 48:

$$\text{net} = -60 + 22 + 48 = +10 \text{ points, a credit of } \$105{,}000$$

Look first at the two 5% strikes. The put costs 60 points and the call 48. Over three months the index forward sits only about 1% above spot (a 5% interest rate less a dividend yield near 1.5%), which on its own would make the call the dearer of the two, so the gap comes from skew and if anything understates it. This is the same skew Example 12.5 found on a single stock, here with the forward slightly above spot rather than pulled below it by a dividend. A zero-cost collar on this index has to set its call nearer to spot than its put.

The fund is now protected between −5% and −15% on the index (about −5.5% to −16.5% on the portfolio) and capped at +5% on the index, about +5.5% on the portfolio; and it has been paid to enter the structure. What it has given up is protection *below* −15%—precisely the scenario the hedge was ostensibly for. Selling the deep put is part of what pays for the structure, so a budgeted program keeps protection against ordinary declines and gives up protection against catastrophes.

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With the bought put moved to the money, wrapped in an exchange-traded fund and reset once a year, this structure is the **buffer ETF**, or defined-outcome fund: over each period the fund absorbs the first 10 or 15% of the index's loss and caps its gain, and such funds held about $78 billion at the end of 2025. Financial advisers buy them for clients near retirement who want to stay in equities with the first losses absorbed, and who accept the cap, and the losses beyond the buffer, in exchange.

Tail hedging inverts the choice: buy puts 30% out of the money on the index (a fall of about 33% in the portfolio) at, say, 6 points, costing $63,000 a quarter, or about 0.5% a year. The premium is small and the payoff enormous when it comes. The difficulty is that Chapter 19 shows deep out-of-the-money puts carry the *highest* implied volatilities of any options on the surface, so this is the dearest insurance relative to its expected payoff, although its premium is the smallest. A strategy costing 0.5% a year that pays off only in a rare crash must pay a great deal on those occasions to justify itself, and whether it does is genuinely contested.

As Section 3.2's case study warned, an option premium sits in a named budget line every year it pays nothing, and that is part of the appeal of collars, which can be struck at no cost.

Options instead of futures: the margin asymmetry

A commercial hedger faces the same choice, and there the decisive difference is not the premium at all. A futures hedge fixes the price in *both* directions and demands cash on every day the market moves against it. A purchased option (a producer's put, a consumer's call) whose premium is paid up front removes only the unfavorable tail, and owes no variation margin (unless the option itself is margined futures-style, Section 16.5), because the worst outcome for a purchased option is that it expires worthless.

That asymmetry is what Section 3.10's hedging failures are made of. Every one of them runs through variation margin on a futures leg, and a hedger holding purchased options instead—puts for a producer, calls for a seller of fixed-price supply like Metallgesellschaft—would have faced no margin call in any of them. The premium therefore buys two distinct things: the retained upside, and the elimination of funding risk. Whether the second is worth paying for depends on the firm's access to liquidity, which is exactly the judgment Section 3.10 says firms get wrong.

Writing options is a different matter entirely. A firm that sells options to reduce its hedging cost (selling calls to fund puts, the collar above) takes the margin obligation back on the written leg, along with an exposure on that leg that is not capped. The premium saved is not free, and such a structure's cash flows can come to resemble the futures hedge it was meant to improve on.

12.11 WHAT EACH STRATEGY IS ACTUALLY A VIEW ON

Table 12.2 is the chapter in one place. It records what each structure bets on, what it costs, and what it can lose.

Three observations organize the table.

The time column is almost the volatility column reversed. Positions that are short volatility are helped by the passage of time, and positions that are long volatility are hurt by it. This is no coincidence. Chapter 18 shows that gamma and theta are tied together: a position cannot hold gamma without paying theta, or sell it without collecting theta. So you are paid to wait or you pay to wait, according to whether you are net long or short options.

Almost nothing is a pure directional bet. Only the outright option, the **vertical spreads** of Section 12.2 (spreads whose legs share an expiration and differ only in strike), and the risk reversal have direction as their dominant exposure. Everything else requires the trader to be right about magnitude or timing as well, and a strategy chosen for its payoff shape frequently embeds a volatility view its holder never formed.

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Table 12.2: Exposures of the main structures, with the net premium and the maximum loss per share at the quotes of Table 10.2 (the covered call writes the 50 call; the protective put buys the 45 put).

Strategy	Direction	Volatility	Time	Net premium	Maximum loss
Covered call	mildly up	short	+	credit 1.44	46.86
Protective put	up	long	−	debit 1.89	5.19
Bull call spread	up	mixed	mixed	debit 2.04	2.04
Bear put spread	down	mixed	mixed	debit 2.14	2.14
Butterfly	neutral	short	+	debit 0.77	0.77
Condor	neutral	short	+	debit 1.70	1.70
Straddle	none	long	−	debit 5.35	5.35
Strangle	none	long	−	debit 3.40	3.40
Calendar spread	neutral	term	+	debit	the debit
Risk reversal	up	skew	mixed	credit 0.30	44.70
Ratio spread	mildly up	short	+	debit 1.36	unlimited

The neutral strategies are all short a sharp move. Butterflies, condors, and calendars all profit from stillness, which means they all lose in a sharp move, even the calendar, which Table 12.2 lists as "term" because its far-dated option gains when implied volatility rises. A portfolio of "diversified" neutral strategies is a concentrated short-volatility position, and it will be discovered to be one on the day it matters.

The table also says whose book each structure ends up on. Vertical spreads belong to directional traders who want a view at a known maximum loss, which is also what the position is margined at (Section 12.14). Straddles and strangles are bought around earnings and central-bank meetings by event traders, and delta-hedged by volatility funds; market makers are usually on the other side. Butterflies and condors, which profit from stillness, suit traders who expect a quiet market, and premium-selling funds run condors in the iron form of Section 12.3. Calendars and diagonals are used by term-structure traders and by dealers managing the maturity profile of their volatility exposure, selling options in the nearest expiration at elevated implied volatility against later ones. Ratio spreads are mostly professional: a dealer or hedge fund uses one to reduce the cost of a directional view, and a put ratio spread sells the low-strike puts whose implied volatility the skew has raised.

12.12 STRUCTURED PRODUCTS ARE PACKAGED STRATEGIES

Most people who own an option strategy did not choose it. They bought a product, and the product is a strategy with a wrapper, a fee, and a name.

A principal-protected note promises the return of capital plus participation in an index. It is a zero-coupon bond plus a call.

A reverse convertible pays a high coupon and repays principal in shares if the stock finishes below its initial level (in the common barrier form, only if it also finishes below a lower barrier). It is a bond plus a written put: the coupon above the risk-free rate is the put premium, as Example 12.8 shows.

An autocallable pays a coupon and redeems early if the index is at or above a level on an observation date, and can return less than principal if it breaches a lower one. It is a written **down-and-in put** (a put that comes into existence only if the index falls through the lower level, the barrier) plus a series of digital coupons. The Chinese "snowball" products of Section 35.7 are exactly this structure.

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Example 12.7 Taking apart a principal-protected note

A five-year note returns 100% of principal at maturity plus α times any rise in an index. What can α be?

The issuer must set aside enough to repay principal. At a five-year rate of 4.5%,

$$\text{zero-coupon bond} = 100\,e^{-0.045\times 5} = \$79.85$$

leaving \$20.15 per \$100 to buy call options. If a five-year at-the-money index call costs 25% of notional,

$$\alpha = \frac{20.15}{25} = 80.6\%$$

Subtract a 2% distribution fee and the budget falls to \$18.15, giving $\alpha = 72.6\%$. The 2% fee cost the investor 8 percentage points of participation, because it is taken out of the \$20.15 option budget, a tenth of it, rather than out of the whole note.

Now vary the rate, holding the call's cost at 25%. At a 1% rate the bond costs $100e^{-0.05} =$ \$95.12, leaving \$4.88, and after the fee \$2.88, for participation of 11.5%. At a 5% rate the bond costs \$77.88, leaving \$22.12, and after the fee \$20.12, for participation of 80.5%.

This explains an entire product cycle. Principal-protected notes were nearly impossible to build attractively through the years of near-zero interest rates, and higher interest rates make them attractive to build again. No change in investor preferences is needed to explain it. The option budget is what is left after the bond, and when the bond is expensive there is nothing left.

Example 12.8 Where a 12% coupon comes from

A one-year reverse convertible on a stock at \$100 pays a 12% coupon. If the stock finishes at or above \$100 the investor receives \$100; if below, they receive the shares instead. The one-year rate is 4.5% and a one-year at-the-money put on the stock is worth \$9.00.

Decompose what the investor owns. They hold a bond, they receive the coupon, and they have written a put struck at \$100:

$$\begin{aligned}\text{bond} &= 100\,e^{-0.045} = \$95.60\\ \text{coupon} &= 12\,e^{-0.045} = \$11.47\\ \text{written put} &= -\$9.00\\ \text{total} &= \$98.07\end{aligned}$$

The investor pays \$100 for a package worth \$98.07. The \$1.93 difference is the fee, and it is not disclosed as one.

More important is where the 12% comes from. It is not compensation for credit risk and has nothing to do with the issuer's creditworthiness beyond the bond leg. Of the coupon, roughly 4.5 points are the risk-free rate and the remainder is the premium on a put the investor has sold, less the fee just computed. The high coupon is insurance income, and the investor is the insurer, against a fall in the stock they end up holding.

The product is therefore attractive in the same conditions as the covered call of Example 10.7, and unattractive in the same ones. It performs worst when the stock falls sharply, which is when the investor most wanted the \$100 back and instead receives shares worth considerably less. A useful test for any high-coupon structured product is to work out what option the buyer has written and ask whether they would have written it deliberately.

Case Study 12.1 The covered-call fund boom

Funds that systematically write calls against an equity portfolio (sold as "derivative income" or "buy-write" strategies) grew from a niche into one of the fastest-gathering

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categories in the ETF market during 2022–2024, with assets approaching $100 billion by the end of 2024.

The appeal is a headline distribution yield far above anything available in equities or, for much of the period, in bonds. The mechanics of Section 12.1 explain where it comes from: the fund is short puts, and the yield is insurance premium. Chapter 19 shows the premium exceeds what the options later pay out, on average, so this is a real source of return and not an illusion.

Two things are nevertheless misdescribed. First, the distribution is not income in the sense investors usually mean. It is partly the option premium and partly a return of the investor's own capital, and a fund can keep paying a high distribution while its net asset value falls, because whatever the premium and dividends do not cover is paid out of the fund's own assets. Second, and more important, the strategy gives up the strong markets (Example 10.7). A fund that caps the upside every month will, over a long horizon containing a few large rallies, badly trail the index it writes against—while having delivered exactly the yield it advertised throughout.

The fund has no maturity and promises nothing about what an investor ends up with: it writes new calls every month indefinitely, and its net asset value floats with the index, less the upside it sells and the capital it pays out. These products convert an uncertain total return into a higher, steadier distribution and a lower expected terminal value. For an investor who needs cash flow and can accept that trade, they do something useful. For an investor who reads the distribution yield as income without cost, they are a disappointment that arrives only after several years.

12.13 MANAGING A POSITION AFTER ENTRY

Every profit diagram in this chapter describes a position held to expiration. Most positions are not, and the decisions taken in between determine the outcome at least as much as the structure chosen at entry.

Rolling. A covered call writer whose stock has risen through the strike faces assignment and the loss of the shares. The standard response is to **roll**: buy back the written call and sell another at a higher strike or a later date. Usually this costs money, since the call being repurchased is now in the money.

Rolling once is ordinary position management. Rolling repeatedly is something else. Each roll realizes a loss on the option leg and re-establishes a short call further out, so a writer who rolls to avoid assignment has converted a defined, capped strategy into an open-ended bet that the stock will stop rising—the one thing they have so far been wrong about. The strategy chosen for its conservatism becomes, through individually reasonable adjustments, a persistent short call on a rising stock.

Repairing. A trader holding a long call on which the stock has fallen can sell a higher-strike call against it, converting the position into a spread and recovering part of the premium. The arithmetic is unkind: the stock has fallen, so the higher-strike call is now cheap, and the repair raises very little.

Pre-committing to exits. A stock position can be held indefinitely; an option position cannot. Time decay means "wait and see" is a decision with a price, and Table 12.2 gives its sign for each structure. The discipline is to decide before entry what will close the position (a target, a stop, or a date), because after entry the decay makes every delay look like a reason to keep waiting for the move that has not yet come.

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Adjustments accumulate costs. Each adjustment adds legs, and Section 12.14 shows what legs cost. A butterfly adjusted twice has crossed eight more bid–ask spreads on top of the four at entry, and on the numbers of Example 12.3 the twelve together exceed the position's entire theoretical value. Frequent adjustment is not risk management; past a point it is a transfer to the market maker.

12.14 THE COSTS THAT DECIDE WHETHER A STRATEGY WORKS

Every profit diagram in this chapter is drawn as though the position could be established at a single price. It cannot, and the gap between the diagram and the outcome is usually decisive.

Each leg pays the bid–ask spread. Example 12.3's butterfly lost 38% of its value at entry, and a four-legged condor crosses four spreads. The number of legs multiplies the transaction cost, and the structures with the most attractive reward-to-risk ratios usually have the most legs.

Exit costs the same again. A butterfly held to expiration avoids the exit spread but accepts whatever the stock does. Closing early pays the spread a second time, which on the numbers above would consume three-quarters of the theoretical value.

Legging risk is real. The risk of Section 10.5, that the market moves between fills, is larger here: on a \$0.56 butterfly built from options worth \$4.425, \$2.67, and \$1.475, a move of about 50 cents in the underlying between fills (1% of the stock price) can exceed the entire position value.

Margin treatment varies. A spread whose loss is capped requires margin equal to that maximum loss; a ratio spread with uncovered short options requires the naked-writing margin of Section 10.8. Two structures with similar diagrams can have very different capital requirements, and the more attractive-looking one is often the more expensive to hold.

Short American legs can be assigned. The case study in Section 11.7 showed a supposedly riskless box destroyed by early assignment. A short call is most likely to be exercised just before an ex-dividend date, when the dividend exceeds its remaining time value, the interest and insurance weighed in Example 11.7; a short put, once it is deep in the money (Section 11.9). Either way one leg disappears at a moment chosen by someone else: a bull call spread whose short call is assigned becomes a long call against short stock, a synthetic long put.

Put together, these determine which strategies are practical for whom. A market maker with negligible transaction costs, the portfolio margining of Section 10.8 (margin set on the net risk of the whole account rather than leg by leg), and the ability to trade packages can hold structures that are not practical for a retail trader crossing full spreads on every leg. The correct strategy depends on who is trading it—a question textbooks usually leave implied.

SUMMARY

Every option position is a bet on some combination of direction, magnitude, time, and skew. Choosing a strategy means matching those exposures to a view actually held.

Put–call parity collapses much of the apparent variety. A covered call is a written put plus cash; a protective put is a purchased call plus cash. Where two routes to one exposure exist, the one with fewer option legs is usually cheaper to trade.

Spreads cap the payoff in exchange for a lower cost and are bets on a range rather than on a direction. Butterflies and condors bet on stillness. Straddles and strangles bet on magnitude—but because both the cost and the required move come from the same implied volatility, a straddle is a bet that realized volatility will exceed implied, not that the stock will move.

Calendar spreads bet on time decay differentials and on the term structure of volatility. Risk reversals bet on skew, and are the same structure as the corporate collar. Strategies that are

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neutral on direction are almost all short volatility, so a portfolio of them is concentrated rather than diversified.

Any piecewise-linear payoff can be replicated by a bond, a forward, and calls, with weights equal to the change in slope at each kink. A digital is the limit of a tight spread, priced at $-\partial c/\partial K$, and that is how dealers hedge it.

Structured products are packaged strategies: a principal-protected note is a bond plus a call, a reverse convertible is a bond minus a put, an autocallable is a written barrier put plus digital coupons. Because the option budget is whatever remains after buying the bond, participation rates depend sharply on the level of interest rates, which is why principal-protected notes became scarce in the years of near-zero interest rates.

Transaction costs decide which strategies are practical. Paying the bid–ask spread on every leg can consume a large share of a butterfly's mid-price value, and structures with the best reward-to-risk ratios generally have the most legs and therefore the highest costs. Margin treatment, legging risk, and early assignment on American legs complete the picture, and together they mean the right strategy depends on who is trading it.

FURTHER READING

The OCC's *Characteristics and Risks of Standardized Options* and McMillan's *Options as a Strategic Investment* set out the standard structures and their margin treatment, which changes more often than textbooks do. On covered calls, Whaley (2002) built the buy-write index and reports its first long-horizon record, and Israelov and Nielsen (2015) decompose the strategy into the equity, short-volatility and timing exposures this chapter identifies. Henderson and Pearson (2011) take a retail structured note apart into a bond and options, as Example 12.8 does, and find it priced about 8% above its value; it is the model for doing the exercise on any term sheet. Carr and Madan (2001) state the replication result of Section 12.8 in general, and Breeden and Litzenberger (1978) take it to the butterfly of vanishing width and the risk-neutral density.

PRACTICE QUESTIONS

12.1 Show, using put–call parity, that a covered call has the same profit profile as a written put. State one respect in which the two positions nevertheless differ in practice.

12.2 Using Table 10.2, construct a bear put spread from the 47.50 and 45.00 puts at the quoted bid and ask. Compute the debit, the maximum profit, the maximum loss, and the breakeven.

12.3 Using Table 10.2, price the 47.50/50/52.50 call butterfly at the quoted bid and ask and at mid prices. Compute what execution costs as a percentage of the mid value, both breakevens at the quotes, and where the profitable range lies relative to the \$48.30 stock price.

12.4 Explain why a strangle that costs 36% less than a straddle can require a larger move to break even. What does this imply about "cheap" option strategies generally?

12.5 A five-year principal-protected note is offered when the five-year rate is 3% and a five-year at-the-money call costs 22% of notional. Compute the participation rate with no fee and with a 1.5% fee.

12.6 Identify what each of the following is a bet on, using the four things to bet on named in the chapter's introduction: (a) a calendar spread; (b) a risk reversal; (c) a condor; (d) a covered call.

12.7 Explain why a portfolio containing butterflies, condors, and calendar spreads is not

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diversified.

12.8 A trader establishes a four-legged condor by executing each leg separately. Describe two distinct ways this can cost more than expected, and state how each is avoided.

12.9 A calendar spread sells the near-dated option and buys the longer-dated one at the same strike. Explain why this is neither a directional trade nor straightforwardly a volatility trade, identify the two effects that determine its profit, and state the move in the underlying that hurts it most.

12.10 A risk reversal buys an out-of-the-money call and sells an out-of-the-money put. Explain why its cost is a direct reading of the skew rather than of the level of volatility, and state what a risk reversal that costs nothing to enter tells you about the two implied volatilities.

12.11 A founder holds a large single-stock position and wants to reduce the risk without selling. Describe the collar that achieves this, explain in whose favor the skew works when the collar is on an equity, and state what the shareholder remains exposed to that a sale would have removed.

12.12 A manager hedges an equity portfolio with index puts rather than index futures. State the beta-adjusted contract count, explain what shape the option hedge buys that the futures hedge does not, and explain why the floor it sets on the portfolio is only approximate even when the beta is correct.

12.13 A producer can hedge by selling futures or by buying puts. Compare the two on cost at inception, retained upside, and cash demands over the life of the hedge, and identify which consideration the textbook comparison usually omits.

FURTHER QUESTIONS

12.14 Example 12.3 found that execution consumed 38% of a butterfly's theoretical value. Derive a general expression for the fraction of a multi-leg strategy's value consumed by spreads, in terms of the number of legs and the proportional spread on each. Then use it to explain why market makers can trade structures retail traders cannot.

12.15 A fund writes one-month at-the-money calls against an equity index every month, collecting a premium of 1.5% of the index each time. Construct the sequence of monthly outcomes over a period containing eleven months of +1% and one month of +15%, and compare the fund's total return with the index's. Then explain why a backtest beginning in a calm period will overstate the strategy's appeal.

12.16 Section 12.12 decomposes an autocallable into a written barrier put plus digital coupons. Using that decomposition, explain what happens to the issuer's hedge as the index approaches the knock-in barrier, and connect your answer to the 2024 episode described in Section 35.7.

12.17 A corporate treasurer is offered a zero-cost collar on a large shareholding. Identify what the treasurer is selling, what determines whether the collar can genuinely be struck at zero cost, and under what market conditions the terms will be most attractive.

12.18 Table 12.2 lists the calendar spread as "term" rather than long or short volatility. Explain precisely what exposure this means, construct a case in which implied volatility rises across all maturities and the calendar spread nevertheless loses money, and state what the position is really long.

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